METHOD VALIDATION OF SD BIOSENSOR'S STANDARD™ F PLATFORM FOR THE RAPID DETECTION AND DIFFERENTIATION OF DENGUE NS1 ANTIGEN, DENGUE IgM/IgG, ZIKA ANTIGEN AND ZIKA IgM/IgG ANTIBODIES IN HUMAN PLASMA AND WHOLE BLOOD SAMPLES

A Research Paper
Submitted in Partial Fulfilment of the Requirement for the Degree of
Master of Science in Medical Microbiology

Of
The University of the West Indies

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2019

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ABSTRACT

Method Validation Of SD Biosensor's STANDARD™ F Platform For The Rapid Detection And Differentiation Of Dengue NS1 Antigen, Dengue IgM/IgG, Zika Antigen and Zika IgM/IgG Antibodies In Human Plasma And Whole Blood Samples

Blair Steele

This retrospective observational study attempted is to evaluate SD BIOSENSOR's STANDARD™ F Fluorescent Immunoassay for the rapid detection and differentiation of Zika virus (ZIKV) Antigen/IgM/IgG and dengue virus (DENV) IgM/IgG and Non-Structural Protein 1 (NS1) antigen. These arboviruses share factors that complicate definitive diagnosis, in light of their tendency to cause concurrent arboviral infections in this region. Whole blood/serum/plasma samples – previously tested by reference/gold standard tests (Enzyme-linked Immunosorbent Assay - ELISA) and Polymerase chain reaction - PCR) were retested with the SD Biosensor STANDARD™ F Fluorescent Immunoassay. Samples were selected between January 2016 and March 2019, from the Laboratory Information System (LIS) at the Virology Laboratory, Department of Microbiology University of the West Indies. The dengue IgG test component of the SD Biosensor Standard™ F kit had the highest individual positive percent agreement (83.33%) when compared to the reference test (ELISA) for the diagnosis of dengue in Jamaica. It was also the most accurate of the SD Biosensor STANDARD™ F tests. Zika IgG detection using this platform had the lowest (0%) individual positive percent agreement. Zika antigen test showed a positive percent agreement of only 2.17%. This is stark contrast to the proven accuracy, sensitivity and specificity provided by the gold standard tests currently in use at this facility. Although the SD Biosensor Standard™ F was indeed able to detect viruses more rapidly, it failed to prove its efficiency as an alternative to the use of gold standard tests in this setting.

Keywords: Zika; Dengue; Sensitivity; Specificity; Immunoassay; Fluorescence; Serology; Rapid; Validation; IgG; IgM; NS1; Antigen; Antibody

ACKNOWLEDGEMENTS

I would like to express my sincerest gratitude to the SD Biosensor Company for providing all the diagnostic kits free of charge.
I would also like to thank the local representative from said company, who took time, in her off-time, on her birthday, to demonstrate the use of the SD Biosensor Standard™ F F100 and F200 machines.

I would also like to thank my supervisor, Dr. Glendee Reynolds-Campbell for her unwavering support.

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Glossary

Word	Definition
Accuracy/Overall Percent Agreement	The accuracy of a measurement is defined as the closeness of the measured value to the true value.
Acute Sample	An Acute sample is defined as that which is collected within the first 5-7 days post onset (DPO) of symptoms,
Antibody	Proteins produced by the immune system in response to antigens (toxins and foreign substances).
Antigen	A molecule (including toxins and foreign substances) that stimulates an immune response (i.e. production of antibodies against said toxin/foreign substance).
Arbovirus	Arthropod-borne viruses (“Arboviruses”) are a group of viruses transmitted by Arthropods (e.g. ticks and mosquitos).
Concurrent	Happening simultaneously.
Convalescent Sample	A convalescent sample is defined as that which is collected 2-4 weeks D.P.O. of symptoms.
Cross-reactivity	When an immune response (production of a particular antibody) raised against a specific antigen is induced by a different antigen.
Endemic	A disease that exists permanently in a particular region or population example Malaria in parts of Africa.
Epidemic	The breakout and rapid spread of an infectious disease in a given

	population over a short period of time.
Flavivirus	Flavivirus is a genus of viruses in the family *Flaviviridae*. This genus includes the West Nile Virus, Dengue Virus, Tick-borne Encephalitis Virus, Yellow Fever Virus, Zika Virus and several other viruses (Wikipedia.com).
Fluorophore	Fluorescent chemical compound that can re-emit light upon light excitation (Wikipedia.com).
In Vitro	Scientific procedures performed outside of the body, that is, within a controlled, lab setting.
In Vivo	"In the living body". An in vivo experiment is one that is done in the body of a living organism, as opposed to in vitro, which is one that is performed outside of the body.
NS1 Protein	NS1 is a 55 kDA glycoprotein secreted by DENV infected cells in vitro and in vivo (Hang et al., 2009). The Dengue NS1 protein can be detected as early as Day 1 post-onset of illness, and may even be detected up to Day 9 post-onset of illness, following seroconversion (Fry et al., 2011). These results suggest that detection of NS1 can be a useful diagnostic tool for early, rapid and accurate diagnosis of Dengue.
Multiplex Assay	An assay that can detect more than one biomarker simultaneously.
Primary Infection	A primary infection is the first time a person is infected by a pathogen. IgM appears first and normally elicits an immune response by days

	5-7 after onset of illness, hence serological methods are not effective until around days 5-7 after clinical onset. Primary and secondary arbovirus infections are differentiated by IgM and IgG serology (Shu & Huang, 2004).
Sample Stability	The integrity of the proteins (antibodies and antigenic components) of the virus(es) in patient specimen (whole blood/serum/plasma).
Secondary Infection	A secondary infection, for example, with dengue, is characterized by the earlier onset of high titres of cross-reactive IgG antibodies before or simultaneously with the IgM responses (Shu & Huang, 2004).
Sensitivity/Positive Percent Agreement	Also known as the “True Positive” refers to the ability of a diagnostic test to correctly identify a positive as actually positive. For example, it is the ability of a diagnostic test to correctly identify people who have a given disease.
Serology	Study of serum and other body fluids (Wikipedia.com).
Serotype	A serologically distinguishable strain of a microorganism. (Wikipedia.com).
Specificity/Negative Percent Agreement	Also known as the “True Negative” refers to the ability of a diagnostic test to correctly identify a negative as actually negative. For example, it is the ability of a diagnostic test to correctly identify people who don’t not have a given disease.
Standard Deviation	A measurement of how much a group of values differ from the mean value of the said group.

Triplex Assay	An assay that can detect three (3) biomarkers simultaneously.
Validation	The process that determines the fitness of an assay, which has been properly developed, optimized and standardized, for an intended purpose. Validation includes estimates of the analytical and diagnostic performance characteristics of a test (OIE Terrestrial Manual, 2010).
Viremic	Viral particles are present in the bloodstream.

List of Abbreviations and Acronyms

Acronym/Abbreviation	Definition
CDC	Centers for Disease Control and Prevention
CHIKV	Chikungunya Virus
DENV	Dengue Virus
DHF	Dengue Hemorrhagic Fever
DPO	Days Post Onset
DSS	Dengue Shock Syndrome
EDTA	Ethylenediaminetetraacetic acid
ELISA	Enzyme-linked Immunosorbent Assay
FIA	Fluorescent Immunoassay
FITC	Fluorescein Isothiocyanate
ICH	International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use.
IFA	Immunofluorescent Assay
IgG	Immunoglobulin-G
IgM	Immunoglobulin-M
JEV	Japanese Encephalitis Virus
LCD	Liquid Crystal Display
LIS	Laboratory Information System
LOD	Limit of Detection
NPA	Negative Percent Agreement
NS1	Non-structural Protein 1
OPA	Overall Percent Agreement
PCR	Polymerase Chain Reaction
PPA	Positive Percent Agreement/Value
PRNT	Plaque-Reduction Neutralization Test
RFU	Relative Fluorescence Units
RT-PCR	Reverse-Transcription Polymerase Chain Reaction
SLEV	St. Louis Encephalitis Virus
UHWI	University Hospital of the West Indies
WHO	World Health Organization
WNV	West Nile Virus
YFV	Yellow Fever Virus
ZIKV	Zika Virus

Introduction

Background

The Zika virus (ZIKV) was first isolated from the blood of a sentinel monkey in the Zika forest, Uganda in 1947. However, until an outbreak in Yap, Micronesia in 2007, ZIKV infections in humans were rare, and originally thought to be as a result of infection with the dengue virus (DENV). DENV has established endemicity in the Caribbean region for several decades, but was only first described in Brazil in 1985 (Charrel et al., 2016). According to the WHO, the major 2016 dengue outbreak in Jamaica was primarily characterized by dengue serotypes 3 (DENV3) and 4 (DENV4). In 2019, laboratory tests have confirmed that DENV3 is the primary serotype currently circulating in Jamaica (WHO, "Dengue fever – Jamaica", 2019).

Both DENV and ZIKV belong to the *Flaviviridae* family. Around 50 to 100 million cases of dengue are reported each year, and around 3 billion people are at risk of infection. Epidemics have been reported in over 100 countries (Fry, et al., 2011).

The dengue, Zika and chikungunya viruses all share similar presentations in early clinical stages. Their infections can be asymptomatic or present with mild early non-specific symptoms such as fever, a pruritic maculopapular rash, conjunctivitis, arthralgia, and myalgia, headache, malaise and fatigue. Much more serious complications are seen in the form of Dengue Hemorrhagic Fever (DHF), Dengue Shock Syndrome (DSS), Zika virus Associated Microcephaly in newborns, as well as other neurologic sequalae such as Guillain Barré Syndrome. This poses a significant diagnostic challenge that is further compounded by concurrent epidemics, geographical overlapping and co-infection. As a result, differential and timely diagnosis to identify the etiological agent is crucial to treatment.

Dengue surveillance, control and case management in endemic countries are also hindered by lack of technical expertise, research capacity and laboratory infrastructure (Guzman & Kouri, 2004).

Serological methods have long been the go-to for arbovirus diagnosis. Primary and secondary arbovirus infections are differentiated by Immunoglobulin-M (IgM) and Immunoglobulin-G (IgG) serology. A primary infection is the first time a person is infected by a pathogen. IgM appears first and normally elicits an immune response by days 5-7 after onset of illness (see Figure 2a.). Hence serological methods are not effective until around days 5-7 after clinical onset. Presence of IgM indicates a recent, ongoing infection. IgM peaks around week 2 post-infection and can persist for 1-3 months. IgG appears at around days 10-14 and usually remains for years if not a lifetime. Serological tests therefore allow for a longer diagnostic window than molecular tests (Figure 1). Serological diagnosis of arboviral infections is maximized by parallel testing for IgM and IgG. For the detection of IgM antibodies, a single serum sample may be used. However, IgG responses require the use of paired acute-convalescent serum to ascertain a 4-fold rise indicative of recent infection. An Acute sample is defined as that which is collected within the first 5-7 days post onset (DPO) of symptoms, whereas a convalescent sample is defined as that which is collected 2-4 weeks DPO of symptoms.

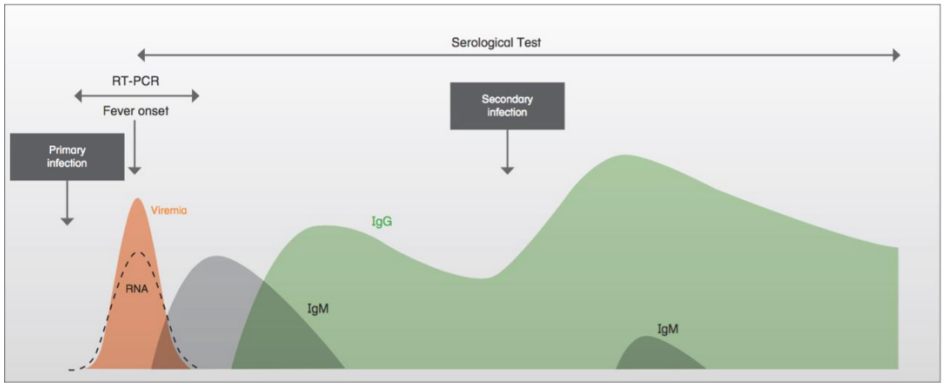


Figure 1. Comparison of the lengths of diagnostic windows of Serological Tests to Molecular tests like RT-PCR

*https://www.cdc.gov/zika/pdfs/denvchikvzikv-testing-algorithm.pdf

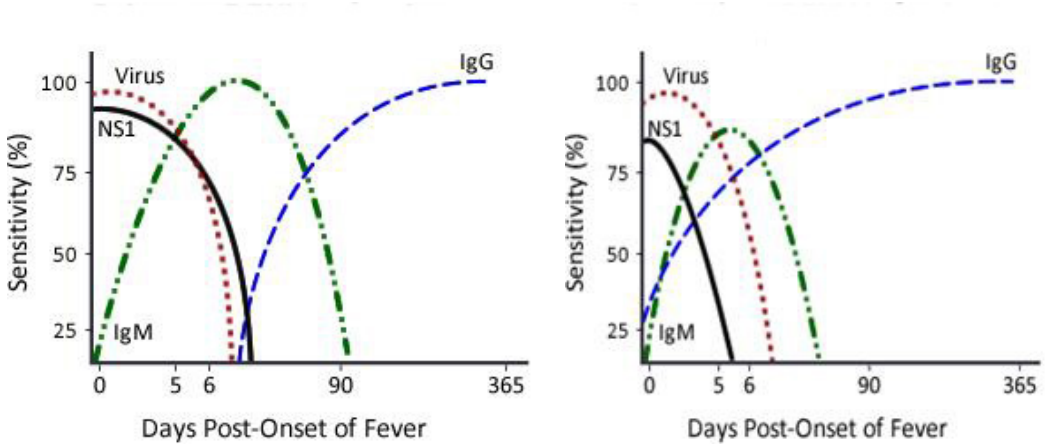
Secondary infection with Dengue is characterized by the earlier onset of high titres of cross-reactive IgG antibodies before or simultaneously with the IgM responses (Shu & Huang, 2004) (Refer to Figure 2b).

The diagnostic challenges, as described earlier, are even further compounded by serological cross-reactivity. This cross-reactivity is likely due to [concurrent] endemic circulation or lately, due to vaccination (Charrel et al., 2016). The greatest challenge with serological cross-reactivity is seen in patients who have been previously vaccinated with flavivirus vaccination or have been exposed to heterologous flavivirus infection (Petersen, Jamieson, & Powers, 2016).

DENV and ZIKV, being from the same family, share similar serological markers that many current IgM-antibody assays cannot differentiate, causing erroneous results (Haug, Kieny, & Murgue, 2016).

As a result, these serological results should be confirmed with other tests using other identifying markers. Such tests include enzyme-linked immunosorbent assays (ELISA), immunofluorescence assays (IFA), and plaque reduction neutralization test (PRNT) (Charrel et al., 2016) (Refer to Figure 3).

According to Charrel et al. 2016, The Euroimmun company is currently the only manufacturer providing detailed validation data that clearly addresses the above-mentioned issue in flavivirus cross-reactivity. The Euroimmun™ Indirect Immunofluorescence Test also had high performance (96% accuracy) (Johnson, Russell, & Goodman, 2016).



A. Primary Dengue Infection **B.** Secondary Dengue Infection

Figures 2. Distinction of Primary and Secondary Dengue Infection. https://www.cdc.gov/dengue/clinicallab/laboratory.html

Relatively inexpensive, rapid Point-Of-Care devices are now being widely used in endemic settings (Fry et al., 2011). Other tests such as neutralization tests are the most specific and are often used to confirm ELISA and FIA results but have had only limited validation (Charrel et al., 2016).

Virus isolation via mosquito cell lines, coupled with immunofluorescence, has contributed significantly to the early diagnosis of the Dengue. However, this technique is very time-consuming and may only be limited to the 5-day window of viremia post-onset of infection (Fry et al., 2011).

A study by Young et al. in 2000 involved the use of monoclonal antibodies in an antigen capture ELISA to detect high levels of dengue virus protein NS1 in the sera of infected patients. NS1 is a 55 kilodalton (kDA) glycoprotein secreted by DENV infected cells in vitro and in vivo (Hang et al., 2009). The dengue NS1 protein can be detected as early as Day 1 post-onset of illness, and may even be detected up to Day 9 post-onset of illness, following seroconversion (Fry et al., 2011). These results suggest that detection of NS1 can be a useful diagnostic tool for early, rapid and accurate diagnosis of dengue.

Whereas a positive NS1 test can confirm a dengue diagnosis, a single, positive IgM and IgG blood sample is only suggestive of a dengue Infection. This is because IgM and IgG antibodies remain detectable in the blood for months. Therefore, in order to serologically confirm an acute Dengue infection, an IgM seroconversion or a four-fold increase of IgG antibody titers in paired sera must be demonstrated. The Centers for Disease Control and Prevention (CDC) reports that IgM tests can be useful for surveillance and support diagnosis of DENV infection in conjunction with clinical symptoms, medical history, and other epidemiologic information. Because IgM persists for >60 days, IgM assays should not be used in dengue-endemic countries as confirmatory tests for current illness. Presence of IgM indicates that a dengue infection has occurred in the past 2–3 months. NS1 sensitivity subsequently decreases over time, and as a result, NS1 tests taken outside of the Day 1 to Day 9 post onset of illness window, may not be useful in dengue virus detection. Inversely, IgM sensitivity is expected to increase as the days increase post onset of illness. Blacksell et al., 2011 confirm this in their study.

In a 2016 study by Steinhagen, Probst, and Radzimski, it was revealed that the NS1-based anti-ZIKV ELISA is a sensitive and highly specific tool for the serodiagnosis of ZIKV infections, eliminating cross-reactions with antibodies to DENV and other flaviviruses. Cabral-Castro Cavalcanti et al. 2016 suggest that laboratory confirmation of Dengue infection in areas of ZIKV and DENV co-transmission should not employ Anti-DENV IgM/IgG reactivity without NS1 detection. Also, of note is that a negative NS1 assay result does not exclude Dengue as a diagnosis (Hang et al., 2009). In a study by Fry et al., 2011, optimal diagnostic sensitivity for acute Dengue was achieved when NS1 testing was combined with IgM and IgG detection. Specificity remained unaffected, and an extended window of detection was also achieved.

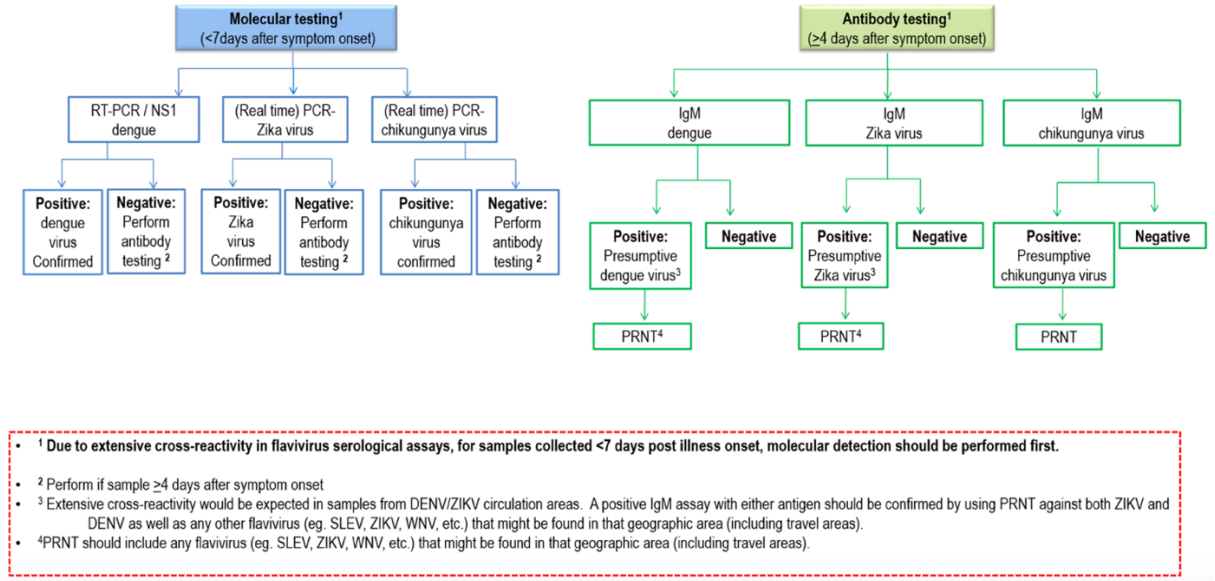


Figure 3. Testing Algorithm for the detection and differential diagnosis of DENV, ZIKV and CHIKV
https://www.cdc.gov/zika/pdfs/denvchikvzikv-testing-algorithm.pdf

The basis of this algorithm put forward by the CDC for the detection and differentiation of DENV and ZIKV infections (see Figure 3) depends on timing of specimen collection (Johnson et al., 2016). At the CDC, samples collected <6 days after onset of illness are first tested by real-time RT-PCR; samples collected on or after day 6 of illness and samples with negative real-time RT-PCR results are tested by the MAC-ELISA. Positive MAC-ELISA results are confirmed by the 90% PRNT (Johnson et al., 2016). PRNT is the most specific serological tool for the determination of arboviral antibodies. (Guzman & Kouri, 2004).

The new alternative and gold standard for Dengue diagnosis is now RT-PCR. RT-PCR allows a wider window of detection, serotyping and quantitation of viral load (Fry et al., 2011). Molecular diagnosis of arboviruses can be done on a wide range of samples including Ethylenediaminetetraacetic acid (EDTA), urine, plasma, saliva, whole blood and serum. Molecular diagnosis can be used to detect ongoing illness and for disease surveillance. Molecular techniques are more expensive, require experienced staff, specialized equipment, and

standardized protocols (Fry et al., 2011). This can be a problem in countries of lower socio-economic status. It should also be noted that short viremic phases (approximately 7 days) complicate the detection of viral RNA and viral isolation, limiting the effectiveness of molecular tests to only the first week after the onset of symptoms as the patient has to be viremic at the time of sample collection (Cabral-Castro, Cavalcanti, & Peralta, 2016). Serology would then have to be employed as a confirmatory test post the viremic phase (Haug et al., 2016).

A myriad of single and multiplex real-time RT-PCR assays for the detection of Arboviruses have been reported. Pabbaraju et al. (2016) developed and validated a triplex real time reverse transcription PCR assay using hydrolysis probes targeting the non-structural protein 5 (NS5) region of ZIKV, non-structural protein 4 (nsP4) from CHIKV, and 3' untranslated region (3'UTR) of DENV1-4". Pabbaraju et al. (2016) have also claimed that the assay can provide conclusive results in less than one hour of run time.

The STANDARD™ F tests from SD Biosensor (SD Biosensor Inc., Suwon, South Korea) aims to allow for a more cost-effective approach to serological diagnosis and differentiation of Dengue NS1, Dengue IgM/IgG, Zika Antigen and Zika IgM/IgG on human serum, plasma and whole blood. Testing may be done on samples collected from day 5-7 of illness and onwards. It utilizes lateral flow, fluorescent immunochromatography with a Europium-based dye as the detection conjugate. See Figure 4 for the principle of Lateral Flow Immunochromatography. The test strips are read by a machine which automates the process, leaving less room for subjectivity. The method also does 2-dimensional reading of barcodes associated with each labeled strip to allow for reduced errors in the manual transfer of patient data. Results, strip expiration dates and lot numbers can be programmed to deliver results to a laboratory or hospital information system, making results available for viewing in real time. Results can also be sent by Bluetooth, direct cable connections or to a data cloud. All components and cartridges/ strips can be kept at room

temperature. Test results are available by this method within a total of 15 minutes (10-minute set up and incubation, and 5 minutes for analysis of cartridge in machine). A total of 240 tests can be read in an hour with batched incubation.

Immunofluorescence is a common laboratory technique that involves the labeling of the said specific recombinant antigen with a fluorophore (a fluorescent dye) like fluorescein isothiocyanate (FITC). There are two main types of immunofluorescence techniques depending on the antibody that is labeled. In direct immunofluorescence the primary antibody to the target antigen is labeled with the fluorophore, whereas in indirect immunofluorescence the primary antibody is unlabeled, and a secondary, labeled antibody is directed against the primary antibody (Figure 5). We are then able to visualize the distribution of fluorophore-labeled antigens under a fluorescent microscope.

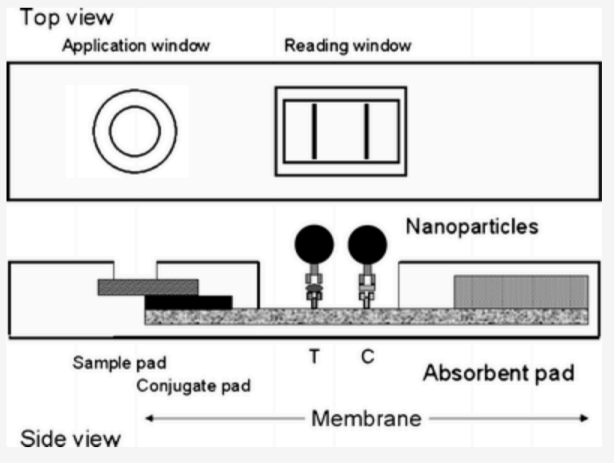


Figure 4. Principle of Lateral Flow Immunochromatography (Posthuma-Trumpie, Korf, & Amerongen, 2008)

Summary of basic principle of Lateral Flow Immunochromatography:

1. A Europium-coated molecule is immobilized at the conjugate pad (detection conjugate). Europium is a fluorescent dye.

2. Patient specimen (+ dilution buffer in some scenarios) is applied to the sample well.
3. If target antibody is present in the patient specimen, it will form a complex with the Europium-conjugated molecule, immobilized on the conjugate pad.
4. This complex then migrates across the test strip (analytical/nitrocellulose membrane), via capillary action, where primary, labeled-antibodies capture said complexes (this is validated by the formation of a control line).
5. The amount of formed dye (i.e. the intensity of the fluorescence) in this analytical membrane area is proportional to the concentration of infectious viral particles in the patient specimen.
6. This intensity is measured by reflectance photometry in STANDARD™ F analyzer. The measured value takes into account the signal strength of the light, and the previously measured blank value.
7. Excess sample is absorbed on absorbent pad.
8. Result is then displayed on device screen, or printed.

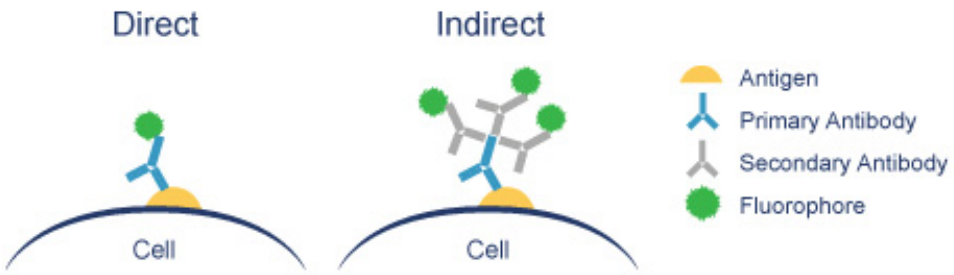


Figure 5. Distinction between Direct and Indirect Immunofluorescence.

http://www.abcam.com/secondary-antibodies/direct-vs-indirect-immunofluorescence

Immunofluorescence provides high specificity and sensitivity in *in vitro* diagnostics. Direct immunofluorescence provides a single target/signal of detection, whereas indirect immunofluorescence provides multiple targets of detection due to the presence of multiple labeled secondary antibodies. Despite the magnitude of the signal being detected, immunofluorescence provides high PPA, i.e. a high rate of detection of the fluorophore, even if it has a low expression/signal of detection. In immunofluorescence, subcellular anatomy and locations of targeted antigens may be visualized. Cell expression may also be observed.

Another advantage of immunofluorescence is that it yields results rapidly. The SD Biosensor platform does this in 5 minutes. The STANDARD™ F platform consists of a fluorescent immunoassay/lateral flow strip and an automated analyzing device into which it the strip with the patient sample can be inserted, initiated and prepared, processed and read. According to SD Biosensor, samples with high titers can yield results from the analyzer in 1.5 minutes. When processing is completed, results are displayed on the analyzer's LCD (Liquid Crystal Display) screen in Relative Fluorescence Units (RFU) and may also be printed on paper. Cut-off indices will determine a positive or negative result. SD Biosensor data suggests the platform has between 97% and 100% sensitivity and specificity for all three arboviruses. Table 3 below shows a comparison of the diagnostic performance of the individual assay parameters to the reference/gold standard methods

Differential diagnoses of arboviruses are particularly important in the midst of concurrent, overlapping circulation and co-infection. Since definitive clinical diagnosis based on symptoms and epidemiological data only for ZIKV and DENV infections is challenging, and complications especially in the pregnant population can be so devastating, it is of utmost importance that testing platforms have the capability of producing accurate results in a timely fashion. A triplex diagnostic tool for simultaneous testing that is capable of eliminating the cross-reactivity issue that occurs with serological tests, and hence accurately differentiates species of

arboviruses is required for correct diagnosis of such infections amongst special populations (travelers returning from endemic areas, children and pregnant women) (Pabbaraju et al., 2016).

Before new testing platforms can be introduced, careful consideration needs to be taken as it pertains to the experience of clinicians; the preparedness of diagnostic laboratories and resources (e.g. reagents); diagnostic testing capacity, sample size, and the limitation of diagnostic laboratories when it comes to emerging diseases (Charrel et al., 2016).

It is with this in mind that this study seeks to validate SD Biosensor's STANDARD™ F platform as a rapid, cost effective, sensitive, specific and accurate serological diagnostic tool useful in our resource limited setting for the detection and differentiation of Dengue NS1, Dengue IgM/IgG and Zika IgM/IgG in human plasma and whole blood samples.

The results will be used to calculate estimates of Positive Percent Agreement and Negative Percent Agreement which will be reported with two-sided 95% confidence intervals. These results should be presented as the % agreement with clinical truth. The study will therefore assess the capacity for the SD Biosensor platform to show similar results to standard tests and therefore answer the question as to whether or not this new platform can offer to the laboratory a more rapid, accurate and differential diagnosis of these arboviral infections.

Materials and Methods

This retrospective descriptive analysis will include the use of plasma and/or whole blood samples that had been previously tested by reference/gold standard PCR and ELISA tests. These reference serological tests include; Euroimmun™ (ZIKV IgM) and SD Bioline™ or Focus™ (DENV IgM) and molecular tests (PCR): Roche™ LightMix® (DENV) and Lanciotti SinglePlex™ (ZIKV) (the gold standard test for ZIKV Antigen). Samples that tested positive or negative for Zika or dengue viruses using the reference/gold standard laboratory testing platforms were retested by the SD Biosensor STANDARD™ F Fluorescent Immunoassay.

Each patient's demographics and previous test results were encoded by a numbering system, and all individually identifying markers removed for blinding of the other investigators. All identifying information was stored in a locked cabinet which only the primary investigator has access to. Samples were randomly selected between January 5, 2016 and March 21, 2019, from the Laboratory Information System (LIS) at the Virology Laboratory, Department of Microbiology University of the West Indies. The samples that were previously tested came from patients who had shown signs or symptoms of a possible arboviral disease. These were patients of the University Hospital of the West Indies (UHWI) and of private physicians. Some of the samples previously tested at the UHWI came from the National Public Health Lab, which receives samples from clinics across the island of Jamaica.

The study was conducted at the Virology laboratory in the Department of Microbiology of the UHWI after ethical approval was obtained (ECP39 17/18). The virology laboratory is the reference laboratory for testing arboviruses in Jamaica and receives specimens from all 14 parishes of the island (Vickers et al., 2015).

Patient consent was obtained at the time of original collection of samples for arboviral screening during their illness.

These samples were stored at -70°C to maintain integrity of the proteins (antibodies and antigenic components) of the viruses.

Sample Size Calculation

The Virology laboratory switched testing platforms between 2016 and 2019. As a result, the numbers of each type of sample tested were affected by the number of samples available for testing by some of the platforms. Previous research using these same samples also affected the volume of sample that remained for testing using the SD Biosensor S Standard™ F platform.

Study selection criteria

1. Plasma, venous whole blood or serum: Stored in a -70ºC refrigerator.
2. Samples must have been tested with the reference/gold standard methods (Positive or Negative)

Exclusion criteria

1. If residual sample was not appropriately stored.
2. If residual sample is suspected to be contaminated with microorganisms.
3. If a volume of residual samples is not enough to proceed with test.

Table 1. Sample Size Formula/Calculation

Sample size = $\frac{(\text{Z Score})^2 \text{x Std Devx (1-Std Dev)}}{(\text{margin of error})^2}$
True Sample size = $\frac{(\text{Sample size x population})}{\text{Sample size+ population-1}}$ $= \frac{((1.96)^2 \text{ x } 0.5(0.5))}{(0.05)^2} = 385$
True Sample size E.g.: ZikV IgM positives $\frac{(385 \text{ x } 50)}{(385+ (50-1)} = 44$

Diagnostic tests

The method for diagnosing Dengue NS1 antigen, as provided by the SD Biosensor Dengue NS1 kit instructions, required placing 100µL of the patient specimen (serum, plasma or whole blood) on an appropriately labeled test device. In the Read-only test mode, up to 240 qualitative tests can be analyzed in an hour. In this scenario, multiple specimens, each applied to the wells of appropriately labeled test devices, are processed consecutively in 15-second intervals. In this read-only test mode, up to 40 qualitative test were run simultaneously during this study. After the samples were applied, the test strips were left on the bench to incubate for 15 minutes, after which they were inserted into the analyzer one by one, and then analyzed simultaneously, in 15 second intervals.

The method for diagnosing Zika antigen was the same – a 100µL of the patient specimen was added to the test device. For diagnosing Dengue and Zika IgM/IgG, a 10µL of the patient specimen was added to the test device.

The device uses an optical sensor to measure fluorescence, at room temperature, between wavelengths of 250-850 nm, in Relative Fluorescence Units (RFUs). When the specimen is applied, it starts an enzymatic reaction which results in release of fluorescence due to activation of a dye (Europium-based) and the amount of formed dye is proportional to the concentration of infectious agent in the specimen. This is the principle behind Lateral Flow Immunochromatography (Figure 4). The intensity of visual color or fluorescence can be measured once detected by reflectance photometry in the STANDARD™ F analyzer. The measured value takes into account the signal strength of the light, and the previously measured blank value. When processing is completed, results are displayed on the analyzer's LCD screen and/or printed on paper. STANDARD™ F uses proprietary algorithms to analyze the data, interpret the signals and determine the test results. The device calculates cut-off indices (thresholds) that correspond with a positive or negative result.

Figures 6a and 6b are examples of results yield on the analyzer's LCD screen (Figure 6a.) and results printed out from the device (Figure 5b.) Each test result is displayed on the LCD and printed out in 15 seconds intervals in cases of simultaneous testing. Cut-off indices (as pre-determined by manufacturers) will determine a positive or negative result. The results will then be compared with that of the previous testing platforms and the positive and negative percent agreements of this new assay determined.

This project is estimated to be completed within 6 months of commencement. Data thus collected and analyzed in this study inclusive of all demographic data collected and stored by the primary investigator will be destroyed within 6 months of completion of the study.

This is a retrospective study of stored samples previously tested for the presence of the said infectious organisms, so patients will experience only minimal risk. The results of this study will also not be used for patient care decisions.



A. Showing test results displayed on the analyzer's screen

B. Showing an example of test results printed out from the device

Figure 6. SD Biosensor Standard™ F analyzer results.



Figure 7. Lab-space set-up with SD Biosensor Machine, Patient Samples, Device Inserts on paper towel with patient samples being incubated for 15 minutes, and a timer

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A. B.

Figure 8. SD Biosensor Standard™ Analyzers. **A.** SD Biosensor Standard™ F F200 machine. **B.** SD Biosensor Standard™ F F100 machine.
http://www.sdbiosensor.com/xe/byBrandStandardF



Figure 9. SD Biosensor Standard™ F kits. **A.** STANDARD™ F Zika Ag FIA. This product performs qualitative analysis on Zika antigen in serum, plasma and whole blood. **B.** STANDARD™ F Dengue NS1 Ag FIA. This product performs qualitative analysis on NS1 antigen of Dengue virus in serum, plasma and whole blood. **C.** STANDARD™ F Dengue IgM/IgG FIA
This product performs qualitative analysis on Dengue virus specific IgM and IgG antibodies in serum, plasma and whole blood.

http://sdbiosensor.com/xe/byBrandStandardF

Data Analysis

Calculating the clinical performances

Evaluation of the clinical performance of the Standard F platform included calculation of its sensitivity and specificity when compared to a gold standard method (i.e. the ZIKA PCR) and the calculation of the positive percent agreement (PPA) and negative percent agreement (NPA) when compared to the laboratory's reference method in use at the time of original testing. To calculate the positive percent agreement (PPA)/sensitivity and negative percent agreement (NPA)/ specificity of an investigational device such as the STANDARD™ F series test kit, test results from the raw data sheet with samples were compared to results obtained from the reference/gold standard methods (PCR and ELISA). The MedCalc™ statistical calculator software (MedCalc Inc., Osten, Belgium), and a table similar to Table 2 below was used to calculate the positive percent agreement (PPA)/sensitivity and the negative percent agreement (NPA)/specificity values. These were calculated and reported along with two-sided 95% confidence intervals. Microsoft Excel (Microsoft Inc., Redmond, Washington, USA) was used for data entry.

Samples with invalid results were retested. Samples that remained invalid after re-testing were excluded from statistical analysis.

Table 2. Two by two table for calculating the positive and negative agreement tests.

Type of sample		Investigational device	
		Positive	Negative
Reference method	Positive	a	b
	Negative	c	d

- **Sensitivity** = a/(a+b) x 100%
- **Specificity** = d/(c+d) x 100%
- **Accuracy (Overall Percent Agreement)** = (a+d)/(a+b+c+d) x 100%
- **a = True Positive**
- **b = False Negative**
- **c = False Positive**
- **c = True Negative**

The results of investigational devices are for performance evaluation only and not for diagnostic purposes.

Results

Patients' samples

The 509 archived samples selected for evaluation were made up of a total of 245 DENV samples and 350 ZIKV samples. Of the 245 DENV samples, 104 were male and 141 were female. The average age of patients was 21.33 years. Of the 350 ZIKV samples, 75 were male and 277 were female. The average age of patients was 26.43 years. Across the study, females made up 418 samples (82.12%) and males made up 179 samples (35.16%) males. The median age across both genders was 23.88 years (range, ≤12 months to 98 years old). The samples were characterized using reference ELISAs for dengue IgM, IgG and NS1 antigen and classified as initially 120 positives (48.97%), and 125 negatives (51.02%). The dengue samples were not classified as acute or convalescent, or primary or secondary. The samples were characterized using reference ELISAs for Zika IgM and IgG and classified as initially 44 positives (28.02%), and 113 negatives (71.9%). The total number of females were 108 (68.78%) and total number of males were 49 (31.21%). A total of 193 samples were characterized using reference PCRs for ZIKV antigen. There was a total of 92 positives (47.66%) and 101 negatives (52.33%). The total number of females were 168 (87.04%) and total number of males were 25 (12.9%).

Overall Diagnostic Performance

The diagnostic performance of the individual assay parameters in comparison to the reference methods is shown in Table 3 below. SD Biosensor Standard F™ published data suggests that the platform has between 97% and 100% sensitivity and specificity for DENV and ZIKV. The dengue IgG test component of the SD Biosensor Standard™ F kit had the highest individual PPA and NPA of 83.33% and 100%, respectively, when compared to the reference test (ELISA) for the diagnosis of dengue in Jamaica. This test component was the most accurate (84.61%). It was

also the most accurate of the SD Biosensor STANDARD™ F tests. The Standard F™ Zika antigen FIA test also had a low sensitivity (2.17%), and a high specificity (100%) with an overall accuracy only 25%. Quite similarly, the Standard F™ Zika IgG FIA test had the lowest (0%) individual sensitivity, but 100% specificity. The Standard™ F Zika IgM/IgG FIA had an accuracy of 52.85%. The SDB-SF Dengue NS1 FIA test component had the second highest accuracy (82.47%), and a PPA and NPA of 63.63% and 99.19%, respectively.

Table 3. Comparison of the diagnostic performance of the individual assay parameters in comparison to the reference/gold standard methods.

SDB-SF FIA test	**PPA/Sensitivity %** (95% CI)	**NPA/Specificity %** (95% CI)	**Overall %** (95% CI)	**Reference/Gold Standard**
DENV IgG	83.33 (51.59 - 97.91)	100 (2.50 - 100)	84.61 (54.55 - 98.08)	Focus™ Dengue IgM (ELISA) Sens = 96%; Spec= 97% SD Bioline™ Dengue IgM/IgG (ELISA) Sens = 94.6% Spec = 96.5%
DENV NS1	63.63 (53.92 - 72.60)	99.19 (95.59 - 99.98)	82.47 (76.99 - 87.12)	100 for both Sens & Spec. Roche™ Lightmix® Dengue (PCR)
ZIKV IgM	2.63 (0.07 - 13.81)	100 (96.73 - 100)	75.17 (67.43 - 81.87)	Sens = N/A; Spec = 98-100% Euroimmun™ Anti-Zika Virus IgM/IgG (ELISA)
ZIKV IgG	0 (0 - 45.93)	100 (15.81 - 100)	25 (3.19 - 65.09)	
ZIKA Antigen	2.17 (0.26 - 7.63)	99.01 (94.61 - 99.97)	52.85 (45.55 - 60.06)	Lanciotti Singleplex™ (PCR); 100% for both Sens & Spec.

CI = confidence interval; Sens = Sensitivity; Spec = Specificity 95% Confidence Interval is shown in parenthesis

Discussion

Differential diagnoses of arboviruses are particularly important in the midst of concurrent, overlapping circulation and co-infection. Since definitive clinical diagnosis based on symptoms and epidemiological data only for ZIKV and DENV infections is challenging, it is of utmost importance that testing platforms have the capability of producing accurate results in a timely fashion, so that the appropriate treatment and public health measure may be implemented.

SD Biosensor Standard F™ (SDB-SF) published data suggests that the platform has between 97% and 100% sensitivity and specificity for DENV and ZIKV (SD Biosensor Product Catalog, 2018). This study, however, failed to achieve results that were comparable in all instances. The gold standard molecular methods require at least an hour or more run time to conduct all the steps from extraction, to analysis and the provision of results. Molecular methods, in general, are also relatively expensive and require several pieces of specialized equipment as well as experienced staff to conduct them. ELISA tests also require more than an hour of run time and experienced staff. Both methods are labour intensive. The SDB-SF platform on the other hand can be completed in a matter of minutes, has a few simple steps, only requires a single piece of equipment and does not share the risks of contamination or other well-known issues that would apply to molecular tests.

During the Zika epidemic that plagued the nation of Jamaica, strict protocol was followed in the testing of samples as directed by the Ministry of Health under the guidance of the CDC and WHO. To be included for PCR testing for presence of Zika antigen samples had to have been collected within the first 5 -7 days of onset of symptoms. All samples collected thereafter were subject to Zika IgM testing. The Zika IgG testing panel was used thereafter to indicate acute infection by determination of a 4-fold rise in IgG titres or indicate previous exposure/ seroconversion. The Standard F™ Zika Antigen FIA test showed a low sensitivity (2.17%), and a

high specificity (100%) with an overall accuracy only 25% compared to the gold standard method. This indicates that that this test was not able to appropriately identify the patients had were currently infected with the virus but did a good job of indicating those that did not have the infection at all. Quite similarly, the Standard F™ Zika IgG FIA test had the lowest (0%) individual PPA, but 100% NPA. This would indicate that this test was better at indicating the patients that were never previously exposed to the Zika virus or those who had yet to seroconvert. For the SD Biosensor Standard™ F Zika IgM/IgG FIA, the run time is 15-20 minutes including incubation—significantly more rapid than the standard method. This test, however, had an accuracy of 52.85%.

The reference method used in the Virology laboratory in the Department of Microbiology of the UHWI for Zika IgM/IgG detection is the EUROIMMUN™ Anti-Zika Virus ELISA. This reference method has a total incubation of approximately 75 minutes, and run-time of 15 minutes – much longer than the SDB-SF platform. A 2016 study by Huzly et al., confirms a 98-100% specificity for EUROIMMUN™ Anti-Zika Virus ELISA (IgG, IgM) (Huzly, Hanselmann, & Schmidt-Chanasit, 2016). The study also confirmed a low cross-reactivity of the said diagnostic kit. This is due to the use of an antigen that is a highly specific, recombinant protein (Huzly et al., 2016). To boost the overall accuracy of the SDB-SF platform for Zika antibody identification, it may be imperative that attempts are made to improve the antigen currently in use in these kits.

The Dengue IgG FIA test component of the SDB-SF platform had the highest individual PPA (83.33%) when compared to the reference test in use at the time (an ELISA). This test also had a NPA of 100% and was the most accurate (84.61% overall agreement). It yielded results in approximately 15-20 minutes, significantly faster than that of the reference method (approximately 90 minutes). A positive Dengue IgG result accompanying a negative IgM result is typically indicative of a past Dengue infection. Vickers et al. (2015) suggest that the lower PPA could be attributed to the possible presence of high IgG antibody titres in those

samples. In that study, the authors suggest that Dengue viral antigens, inclusive of NS1, may form immune complexes with high levels of Dengue IgG antibodies and thus, IgG becomes undetectable, resulting in low sensitivity. The converse is also true as the concept of lower antigen levels correlating with an increase in IgG titers (Andries et al., 2012). A future study should be undertaken to explore the relationship between infection type (primary versus secondary Dengue) and PPA. Blacksell et al., 2011 reported higher sensitivities among patients with secondary DENV infection.

The reference methods used in the Virology laboratory in the Department of Microbiology of the UHWI for Dengue IgM detection, shifted during the 2016-2019 period. The tests used were: (1) Focus™ Diagnostics Dengue Virus IgM Capture DxSelect™ and (2) SD Bioline™ Dengue IgM/IgG, a rapid immunochromatographic diagnostic test. For the Focus™ Diagnostics Dengue Virus IgM Capture DxSelect™, the procedure is approximately 1 hour and 40 minutes including incubation and run-time –significantly slower than the alternative, Standard™ F test. The kit insert suggests this time may be cut down to 60 minutes. This test has a PPA and NPA of 96% and 97%, respectively. The SD Bioline™ Dengue IgM/IgG procedure yields a result in approximately 15-20 minutes and a PPA of 94.6% and NPA of 96.5% according to company data (Abbot Laboratories, Chicago, USA).

In this current study, the SDB-SF dengue NS1 FIA test component had the second highest accuracy (82.47%), and a sensitivity and specificity of 63.63% and 99.19%, respectively. This relatively moderate sensitivity could be due to the time of sampling, that is, whether the sample was collected early or late in the course of disease (Kikuti et al., 2019). Due to a laboratory mishap with SD Biosensor Standard F™ Dengue IgM/IgG FIA kits, sufficient quantities of the test strips were not available to test the DENV IgM samples.

Simultaneously increasing the diagnostic parameters (i.e. increasing the number of biomarkers being detected), will increase the usefulness (sensitivity and specificity) of the kit in detecting the respective arbovirus.

IgM is considered a marker of recent infection and so this limits its usefulness when used alone as a single biomarker in the diagnosis of Dengue infections (Vickers et al., 2015). The Dengue NS1 antigen is found in acute phase (1-2 days after infection), even before IgM is detectable. Vickers et al. 2015 explored different testing strategies by exploring different test combination approaches with the IgM/IgG and NS1 biomarkers and found that combining two or more of the individual components resulted in even greater levels of diagnostic sensitivities (Vickers et al., 2015). Using combination strategy enhances the overall diagnostic performance of the rapid dengue diagnostic kits, and may also do so for the diagnostic kits of other arboviruses. Vickers et al. also combined the IgM parameter with detection of either Dengue NS1 alone or both NS1 and Dengue IgG which significantly improved the sensitivity of their diagnostic kit, without decreasing the specificity (Vickers et al., 2015). dengue and Zika viruses, being from the same family, share similar serological markers that many current IgM-antibody assays cannot differentiate (Haug et al., 2016). The levels of cross-reactivities between the said arboviruses, with other flaviviruses, or with other diseases such as malaria and leptospirosis were not specifically tested for. Brown et al. (2009) showed that among dengue suspected cases in Jamaica, 59% (26/435) had cross-reactive antibodies to West Nile Virus (WNV). Cross-reactivity in dengue NS1 and Zika virus antigen tests would both have diagnostic and medically significant consequences (Matheus et al., 2016). It is therefore recommended that a future study be undertaken to demonstrate the effect of cross-reactive antibodies on the SD Biosensor Standard™ F platform and its ability to differentiate the various organisms. It is recommended that further studies be done to investigate the effect of disease severity and other viral serotypes (DENV-1 to DENV-4) on the performance of the SD Biosensor Standard™ F Kit (Vickers et al., 2015. During the Zika virus outbreak, it was noted that some patients were dually affected with both Zika and dengue viruses.

In this current study, there were 44 samples that tested positive for both DENV and ZIKV by the reference tests (PCR and ELISA). Thirty-four of the Forty-four samples were then retested by the respective SDB-SF FIA test component (Zika IgM/IgG FIA, Zika antigen FIA, dengue IgM/IgG FIA or dengue NS1 antigen FIA). Of the ZIKV reference method positive samples retested, 0/22 came back positive by the SD Biosensor Standard™ F and only 3/22 of the positive DENV samples retested, came back positive. In order to assess cross-reactivity and whether the SDB-SF FIA platform is able to distinguish between DENV and ZIKV, the overall sensitivity of the platform has to be improved.

This study relied on the accuracy of laboratory record keeping of the previous results to which this study will be compared, and accurate documentation of sample storage location. In general, retrospective studies may be subject to biases such as information and selection biases. For dengue, in particular, there could be misclassification bias due to the use of single serum samples, which would not be able to demonstrate dengue seroconversion as may occur with paired samples (Vickers et al., 2015).

The integrity of the result of this study are only as good as the samples used in experimentation. There is a possibility that the protein in some samples may have been degraded during storage, sample preparation or experimentation. Peters et al. (2007) suggests that a full validation of a method must include stability experiments (including freeze/thaw sample stability) for the various stages of analysis including storage prior to analysis (including under storage conditions).

Another limitation was the result of incorrectly logged samples in the LIS. As a result, some samples could not be found. In some instances, the volume of some samples was also insufficient, and as a result, the sample selected had to be randomly replaced.

A cause for concern for this study is that dengue samples were not classified as either primary or secondary infections. A 2010 study by Osorio et al., found a decreased sensitivity of all assays in severe cases

than in non-severe cases. These were more likely to be secondary dengue infections (Osorio, Ramirez, & Bonelo, 2010).

Another weakness of this study is that the diagnostic accuracy of the results presented here may be biased toward secondary Dengue infections, which would be characteristic of a region, like Jamaica, where dengue is endemic. Also, of concern is that the study may have used specimens that may have been biased toward particular dengue serotypes.

Further studies in settings where primary dengue is dominant and serotype identification is done, are required to confirm the diagnostic accuracy of these assays (Blacksell et al., 2011).

The SD Biosensor Standard™ F dengue IgM/IgG FIA also had a limitation i.e. instrumentation error. Extremely thick blood samples could not be run (yielded an *Invalid* result) even after the required dilution. This required reselection and replacement with. Next available sample. Assessments of the "Limits of Detection" (LOD) are therefore necessary. There were a total of 3 invalid samples. The invalid rate, that is, the number of invalid samples divided by the total number samples tested was 0.78%.

It has been shown that the generation of auto-antibodies against platelets, endothelial cells and coagulatory molecules are dengue-associated, and molecular mimicry between these molecules and host proteins (Ainulkhir et al., 2018)

Another factor that could have influenced results is the Cut-Off Index (i.e. the upper and lower limits of detection. This factor influences the accuracies, sensitivities and specificities. A reassessment of the diagnostic performance of updated versions of the SD Biosensor Standard™ F machine and kits, could be done to determine if updated versions of the proprietary Standard™ F algorithms used to analyze the data, have truly improved.

Zammarchi et al., (2019) also investigated the performance of the SD Biosensor Standard™ F FIA to rapidly (≤15 mins) detect DENV IgG, IgM and NS1 Ag, and ZIKV IgG, IgM, and Ag, in Florence, Italy. That study was

able to characterize DENV and ZIKV samples as acute or past infection. For acute dengue infection, the study reported a sensitivity of 100% when the SDB-SF DENV NS1 Ag and/or IgM FIA was combined. In past dengue infection, the SDB-SF DENV IgG FIA had a sensitivity of 70%. The SDB-SF DENV NS1 Ag, IgG and IgM FIA tests had specificities of 87.5%, 83.5%, and 91.7%, respectively. The authors also reported that the SDB-SF ZIKV Ag FIA test did not improve the sensitivity in detecting acute ZIKV infection. They defined a past ZIKV infection as 32-183 days after symptom onset, and reported that, in past ZIKV infection, the SDB-SF ZIKV IgG and IgM FIA tests had a sensitivity of 40% and 80%, respectively. The SDB-SF ZIKV Ag, IgM and IgG FIA tests had specificities of 92.6%, 100%, and 97%, respectively.

In this study of the performance of the SD Biosensor Standard™ F FIA to rapidly (≤15 mins) detect DENV IgG, IgM and NS1 Ag, and ZIKV IgG, IgM, and Ag, in Jamaica, the DENV or ZIKV samples were not classified as acute or convalescent, or primary or secondary. A limitation of this study is that a comparison could not be made between the sensitivities and specificities of acute and past DENV and ZIKV infections. Further studies of the performance of the SD Biosensor Standard™ F FIA in detecting DENV and ZIKV in Jamaica can characterize the samples as such, and compare the sensitivities and specificities between acute and past DENV and ZIKV infections.

Conclusions

SD Biosensor Standard F™ published data suggests that the platform has between 97% and 100% sensitivity and specificity for DENV and ZIKV. Although these kits were more capable of detecting the respective infectious virus more rapidly than the reference/gold standard tests, the results of this study reveal that the SD Biosensor Standard F™ diagnostic kits may only prove to be an ideal alternative to the current reference methods used in the Virology laboratory in the Department of Microbiology of the UHWI, if adjustments are made that improve their overall sensitivity.

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Appendix A

Table A. Summary of study

TITLE	Method validation of SD Biosensor's STANDARD™ F platform for the rapid detection and differentiation of Dengue NS1 antigen, Dengue IgM/IgG, Zika Ag, and Zika IgM/IgG antibodies in human plasma and whole blood samples.
STUDY NUMBER	ECP 39 17/18
MAIN OBJECTIVE	1. To assess the positive percent agreement and negative percent agreement of the STANDARD™ F platform (SD Biosensor Inc., Suwon, South Korea) and gold standard methods for the detection of Zika virus, and Dengue virus IgM/IgG antibodies and Dengue virus NS1 antigen.
STUDY DESIGN	Retrospective observational study, to analyze the agreement of STANDARD™ F test kit results compared with reference methods using leftover specimens in the Department of Microbiology.
PLANNED SPECIMEN SIZE AND TYPE	1. Type of specimen: human whole blood, serum and/or plasma *2.* Sample size: 1764 *A. At least 248 positive and 590 negative clinical specimen tested by gold standard (approved) serological Zika virus IgM and IgG assays and rRT-PCR Zika virus assay.* B. *At least 295 positive and 599 negative clinical specimen tested by gold standard (approved) serological Dengue virus IgM and IgG assays and rRT-PCR Dengue virus assay.*
PRODUCT TO BE EVALUATED	STANDARD™ F Zika IgM/IgG FIA (investigational device) STANDARD™ F Zika Ag FIA STANDARD™ F Dengue NS1 Ag FIA STANDARD™ F Dengue IgM/IgG FIA
STATISTICAL ANALYSIS	Estimates of Positive Percent Agreement and Negative Percent Agreement should be reported with two-sided 95% confidence intervals. These results should be presented as the % agreement with clinical truth.
REFERENCE/ GOLD STANDARD METHODS	Euroimmun™ (ZIKV IgM) and SD Bioline™ or Focus™ (DENV IgM) and molecular tests (PCR): Lanciotti SinglePlex™ (ZIKV) and Roche™ LightMix® (DENV).
STUDY DURATION	Within 10 months after clinical performance study protocol approved by the UHWI/UWI/FMS ethics committee
STUDY INSTITUTION	University of the West Indies

GCP CONFORMATION	This clinical trial will be conducted according to international standards of Good Clinical Practice (GCP) and submitted to a properly constituted Institutional Review Board (IRB), in agreement with local legal prescriptions for formal approval of the trial conduct.

Appendix B – SD Biosensor Standard F™ Results

Table B1. SD Biosensor Standard F™ ZIK IgM - POS Results

	Reference test result	Sd IgM Result	COI
1.	Pos	Negative	0.06
2.	Pos	Negative	0.09
3.	Pos	Negative	0.82
4.	Pos	Positive	1.87
5.	Pos	Negative	0.02
6.	Pos	Negative	0.03
7.	Pos	Negative	0
8.	Pos	Negative	0.01
9.	Pos	Negative	0
10.	Pos	Negative	0.02
11.	Pos	Negative	0.05
12.	Pos	Negative	0.35
13.	Pos	Negative	0.02
14.	Pos	Negative	0.07
15.	Pos	Negative	0.03
16.	Pos	Negative	0.09
17.	Pos	Negative	0.04
18.	Pos	Negative	0.05
19.	Pos	Negative	0
20.	Pos	Negative	0
21.	Pos	Negative	0
22.	Pos	Negative	0.02
23.	Pos	Negative	0.06
24.	Pos	Negative	0.33
25.	Pos	Negative	0.01
26.	Pos	Negative	0.08
27.	Pos	Negative	0.05
28.	Pos	Negative	0.2
29.	Pos	Negative	0.05
30.	Pos	Negative	0
31.	Pos	Negative	0.01
32.	Pos	Negative	0.05
33.	Pos	Negative	0.05
34.	Pos	Negative	0.03
35.	Pos	Negative	0.01
36.	Pos	Negative	0.06
37.	Pos	Negative	0.02
38.	Pos	Negative	0.12

Pos = Positive

Table B2. SD Biosensor Standard F™ ZIK IgM – NEG results

	Reference test result	Sd IgM Result	COI
1.	Neg	Negative	0.04
2.	Neg	Negative	0.07
3.	Neg	Negative	0.04
4.	Neg	Negative	0.05
5.	Neg	Negative	0.06
6.	Neg	Negative	0.04
7.	Neg	Negative	0.01
8.	Neg	Negative	0.06
9.	Neg	Negative	0.14
10.	Neg	Negative	0.01
11.	Neg	Negative	0
12.	Neg	Negative	0.05
13.	Neg	Negative	0.03
14.	Neg	Negative	0.02
15.	Neg	Negative	0
16.	Neg	Negative	0.04
17.	Neg	Negative	0.05
18.	Neg	Negative	0.05
19.	Neg	Negative	0.02
20.	Neg	Negative	0
21.	Neg	Negative	0.05
22.	Neg	Negative	0.01
23.	Neg	Negative	0.02
24.	Neg	Negative	0
25.	Neg	Negative	0
26.	Neg	Negative	0.04
27.	Neg	Negative	0.05
28.	Neg	Negative	0.04
29.	Neg	Negative	0
30.	Neg	Negative	0.02
31.	Neg	Negative	0.03
32.	Neg	Negative	0
33.	Neg	Negative	0.07
34.	Neg	Negative	0.05
35.	Neg	Negative	0
36.	Neg	Negative	0
37.	Neg	Negative	0.09
38.	Neg	Negative	0
39.	Neg	Negative	0.01
40.	Neg	Negative	0

41.	Neg	Negative	0
42.	Neg	Negative	0
43.	Neg	Negative	0.06
44.	Neg	Negative	0.02
45.	Neg	Negative	0
46.	Neg	Negative	0
47.	Neg	Negative	0
48.	Neg	Negative	0.01
49.	Neg	Negative	0
50.	Neg	Negative	0
51.	Neg	Negative	0.03
52.	Neg	Negative	0.01
53.	Neg	Negative	0.02
54.	Neg	Negative	0.05
55.	Neg	Negative	0.04
56.	Neg	Negative	0.02
57.	Neg	Negative	0
58.	Neg	Negative	0
59.	Neg	Negative	0
60.	Neg	Negative	0.01
61.	Neg	Negative	0
62.	Neg	Negative	0.03
63.	Neg	Negative	0.04
64.	Neg	Negative	0.2
65.	Neg	Negative	0.05
66.	Neg	Negative	0.03
67.	Neg	Negative	0
68.	Neg	Negative	0.08
69.	Neg	Negative	0
70.	Neg	Negative	0.33
71.	Neg	Negative	0.03
72.	Neg	Negative	0.03
73.	Neg	Negative	0
74.	Neg	Negative	0
75.	Neg	Negative	0
76.	Neg	Negative	0
77.	NEG	Negative	0
78.	NEG	Negative	0
79.	NEG	Negative	0.01
80.	NEG	Negative	0
81.	NEG	Negative	0.02
82.	NEG	Negative	0
83.	NEG	Negative	0
84.	NEG	Negative	0

85.	NEG	Negative	0.03
86.	NEG	Negative	0
87.	NEG	Negative	0
88.	NEG	Negative	0.05
89.	NEG	Negative	0
90.	NEG	Negative	0.06
91.	NEG	Negative	0.02
92.	NEG	Negative	0.03
93.	NEG	Negative	0.07
94.	NEG	Negative	0.06
95.	NEG	Negative	0.05
96.	NEG	Negative	0.04
97.	NEG	Negative	0.04
98.	NEG	Negative	0
99.	NEG	Negative	0.03
100.	NEG	Negative	0.02
101.	NEG	Negative	0
102.	NEG	Negative	0.03
103.	NEG	Negative	0.04
104.	NEG	Negative	0.01
105.	NEG	Negative	0
106.	NEG	Negative	0.02
107.	NEG	Negative	0.01
108.	NEG	Negative	0.05
109.	NEG	Negative	0
110.	NEG	Negative	0.07
111.	NEG	Negative	0.03

NEG = Negative

Table B3. SD Biosensor Standard F™ ZIKA Antigen – POS results

	Gold Standard		
	ZIK PCR SERUM	SD Result	COI
1.	POS	Negative	0
2.	POS	Negative	0.55
3.	POS	Positive	1.62
4.	POS	Negative	0.04
5.	POS	Negative	0.02
6.	POS	Negative	0.05
7.	POS	Negative	0.04
8.	POS	Negative	0.02
9.	POS	Negative	0.02
10.	POS	Negative	0.02

11.	POS	Negative	0.02
12.	POS	Negative	0.05
13.	POS	Negative	0.05
14.	POS	Negative	0.07
15.	POS	Negative	0.02
16.	POS	Negative	0.11
17.	POS	Negative	0.01
18.	POS	Negative	0.01
19.	POS	Negative	0.03
20.	POS	Negative	0.02
21.	POS	Negative	0.02
22.	POS	Negative	0.02
23.	POS	Negative	0.02
24.	POS	Negative	0
25.	POS	Negative	0.01
26.	POS	Negative	0.04
27.	POS	Negative	0.01
28.	POS	Negative	0.03
29.	POS	Negative	0.02
30.	POS	Negative	0.05
31.	POS	Negative	0.04
32.	POS	Negative	0.01
33.	POS	Negative	0.11
34.	POS	Negative	0.04
35.	POS	Negative	0.03
36.	POS	Negative	0.05
37.	POS	Negative	0.03
38.	POS	Negative	0.18
39.	POS	Negative	0.16
40.	POS	Negative	0.01
41.	POS	Negative	0.03
42.	POS	Negative	0.03
43.	POS	Negative	0.01
44.	POS	Negative	0.04
45.	POS	Negative	0
46.	POS	Negative	0.03
47.	POS	Negative	0.03
48.	POS	Negative	0
49.	POS	Negative	0.04
50.	POS	Negative	0
51.	POS	Negative	0.03
52.	POS	Negative	0.01
53.	POS	Negative	0.01
54.	POS	Negative	0.02

55.	POS	Negative	0.09
56.	POS	Negative	0.02
57.	POS	Negative	0.04
58.	POS	Negative	0.04
59.	POS	Negative	0
60.	POS	Negative	0.02
61.	POS	Negative	0.03
62.	POS	Negative	0.07
63.	POS	Negative	0.02
64.	POS	Negative	0.02
65.	POS	Negative	0
66.	POS	Negative	0.01
67.	POS	Positive	5.63
68.	POS	Negative	0.01
69.	POS	Negative	0.02
70.	POS	Negative	0.02
71.	POS	Negative	0.02
72.	POS	Negative	0
73.	POS	Negative	0.02
74.	POS	Negative	0.1
75.	POS	Negative	0.05
76.	POS	Negative	0.02
77.	POS	Negative	0.01
78.	POS	Negative	0.01
79.	POS	Negative	0.02
80.	POS	Negative	0.02
81.	POS	Negative	0.03
82.	POS	Negative	0.02
83.	POS	Negative	0.02
84.	POS	Negative	0.03
85.	POS	Negative	0.02
86.	POS	Negative	0.02
87.	POS	Negative	0.04
88.	POS	Negative	0
89.	POS	Negative	0.03
90.	POS	Negative	0.03
91.	POS	Negative	0
92.	POS	Negative	0.05

POS = Positive

Table B4. SD Biosensor Standard F™ ZIKA Antigen – NEG results

	Gold Standard		
	ZIK PCR SERUM	SD Result	COI
1.	NEG	Negative	0.02
2.	NEG	Negative	0.33
3.	NEG	Negative	0.03
4.	NEG	Negative	0
5.	NEG	Negative	0.02
6.	NEG	Negative	0.09
7.	NEG	Negative	0.02
8.	NEG	Negative	0.02
9.	NEG	Negative	0
10.	NEG	Negative	0.07
11.	NEG	Negative	0.03
12.	NEG	Negative	0
13.	NEG	Negative	0.02
14.	NEG	Negative	0.01
15.	NEG	Negative	0.19
16.	NEG	Negative	0.02
17.	NEG	Negative	0.14
18.	NEG	Negative	0.02
19.	NEG	Negative	0.05
20.	NEG	Negative	0.01
21.	NEG	Negative	0.06
22.	NEG	Negative	0.02
23.	NEG	Negative	0.01
24.	NEG	Negative	0.01
25.	NEG	Negative	0.01
26.	NEG	Negative	0.01
27.	NEG	Negative	0.05
28.	NEG	Negative	0.02
29.	NEG	Negative	0.02
30.	NEG	Negative	0.02
31.	NEG	Negative	0.16
32.	NEG	Negative	0.02
33.	NEG	Negative	0.06
34.	NEG	Negative	0.02
35.	NEG	Negative	0.1
36.	NEG	Negative	0.01
37.	NEG	Negative	0.02
38.	NEG	Negative	0.02
39.	NEG	Negative	0.02

40.	NEG	Negative	0.08
41.	NEG	Negative	0.02
42.	NEG	Negative	0.01
43.	NEG	Negative	0.02
44.	NEG	Negative	0.03
45.	NEG	Negative	0.1
46.	NEG	Negative	0.02
47.	NEG	Negative	0.02
48.	NEG	Negative	0
49.	NEG	Negative	0.69
50.	NEG	Negative	0.81
51.	NEG	Negative	0.04
52.	NEG	Negative	0.12
53.	NEG	Negative	0.01
54.	NEG	Negative	0.01
55.	NEG	Negative	0
56.	NEG	Negative	0.03
57.	NEG	Negative	0.01
58.	NEG	Negative	0.05
59.	NEG	Negative	0.02
60.	NEG	Negative	0
61.	NEG	Negative	0.04
62.	NEG	Negative	0.01
63.	NEG	Negative	0.13
64.	NEG	Negative	0.15
65.	NEG	Negative	0.02
66.	NEG	Negative	0.01
67.	NEG	Negative	0.01
68.	NEG	Negative	0.04
69.	NEG	Positive	1.21
70.	NEG	Negative	0.17
71.	NEG	Negative	0
72.	NEG	Negative	0.2
73.	NEG	Negative	0.03
74.	NEG	Negative	0.03
75.	NEG	Negative	0.62
76.	NEG	Negative	0.01
77.	NEG	Negative	0.02
78.	NEG	Negative	0
79.	NEG	Negative	0.01
80.	NEG	Negative	0.02
81.	NEG	Negative	0.02
82.	NEG	Negative	0.03
83.	NEG	Negative	0.08

84.	NEG	Negative	0.01
85.	NEG	Negative	0
86.	NEG	Negative	0.01
87.	NEG	Negative	0.33
88.	NEG	Negative	0.06
89.	NEG	Negative	0.01
90.	NEG	Negative	0.03
91.	NEG	Negative	0.02
92.	NEG	Negative	0.01
93.	NEG	Negative	0.03
94.	NEG	Negative	0.01
95.	NEG	Negative	0.03
96.	NEG	Negative	0.07
97.	NEG	Negative	0.01
98.	NEG	Negative	0
99.	NEG	Negative	0.01
100.	NEG	Negative	0.02
101.	NEG	Negative	0.01

NEG = Negative

Table B5. SD Biosensor Standard F™ ZIK IgG – POS results

	Reference test		
	ZIKV IgG result	**Sd IgG Result**	**COI**
1.	POSITIVE	Negative	0.03
2.	POSITIVE	Negative	0.05
3.	POSITIVE	Negative	0.03
4.	POSITIVE	Negative	0.01
5.	POSITIVE	Negative	0.03
6.	POSITIVE	Negative	0.09

Table B6. SD Biosensor Standard F™ ZIK IgG – NEG results

	Reference test		
	ZIKV IgG result	**Sd IgG Result**	**COI**
1.	NEGATIVE	Negative	0.06
2.	NEGATIVE	Negative	0

Table B7. SD Biosensor Standard F™ Dengue IgG – POS results

	Reference test	SD BIOSENSOR	
	IgG Result	IgG Result	COI
3.	Positive	Positive	1.29
4.	Positive	Positive	2.11
5.	Positive	Positive	1.70
6.	Positive	Negative	0.39
7.	Positive	Positive	3.20
8.	Positive	Negative	0.79
9.	Positive	Positive	0
10.	Positive	Positive	3.68
11.	Positive	Positive	1.41
12.	Positive	Positive	2.12

Table B8. SD Biosensor Standard F™ Dengue IgG – NEG results

	Reference test	SD BIOSENSOR	
	IgG Result	IgG Result	COI
1.	NEGATIVE	Negative	0.51

Table B9. SD Biosensor Standard F™ Dengue NS1- POS results

	Reference test result		
	NS1	SD F result	COI
1.	ELISA+	Positive	70.93
2.	ELISA+	Positive	
3.	ELISA+	Negative	
4.	ELISA+	Positive	71.10
5.	ELISA+	Positive	68.70
6.	ELISA+	Positive	
7.	ELISA+	Positive	67.26
8.	ELISA+	Positive	
9.	ELISA+	Positive	
10.	ELISA+	Positive	
11.	ELISA+	Negative	
12.	ELISA+	Negative	
13.	ELISA+	Positive	71.35
14.	ELISA+	Positive	38.99
15.	ELISA+	Positive	71.63
16.	ELISA+	Positive	70.68
17.	ELISA+	Positive	68.39
18.	ELISA+	Positive	70.95

19.	ELISA+	Positive	71.69
20.	ELISA+	Positive	67.66
21.	ELISA+	Positive	2.26
22.	ELISA+	Positive	4.93
23.	ELISA+	Positive	45.99
24.	ELISA+	Positive	33.75
25.	ELISA+	Negative	0.04
26.	ELISA+	Negative	0.04
27.	ELISA+	Positive	69.63
28.	ELISA+	Positive	31.83
29.	ELISA+	Positive	71.50
30.	ELISA+	Negative	0.02
31.	ELISA+	Positive	63.50
32.	ELISA+	Negative	0.06
33.	ELISA+	Positive	21.44
34.	ELISA+	Negative	0.06
35.	ELISA+	Negative	0.02
36.	ELISA+	Negative	0.01
37.	ELISA+	Negative	0.03
38.	ELISA+	Positive	1.68
39.	ELISA+	Positive	70.63
40.	ELISA+	Positive	70.80
41.	ELISA+	Negative	0.45
42.	ELISA+	Negative	0
43.	ELISA+	Positive	58.77
44.	ELISA+	Negative	0.01
45.	ELISA+	Negative	0.11
46.	ELISA+	Positive	2.60
47.	ELISA+	Positive	70.95
48.	ELISA+	Negative	0.05
49.	ELISA+	Positive	2.58
50.	ELISA+	Positive	71.33
51.	ELISA+	Positive	6.45
52.	ELISA+	Positive	38.07
53.	ELISA+	Positive	20.12
54.	ELISA+	Negative	0.05
55.	ELISA+	Positive	70.70
56.	ELISA+	Positive	70.95
57.	ELISA+	Positive	10.19
58.	ELISA+	Positive	69.33
59.	ELISA+	Negative	0.57
60.	ELISA+	Positive	69.57
61.	ELISA+	Positive	51.35
62.	ELISA+	Negative	0.11

63.	ELISA+	Negative	0.42
64.	ELISA+	Positive	69.86
65.	ELISA+	Positive	70.07
66.	ELISA+	Negative	0.05
67.	ELISA+	Positive	37.18
68.	ELISA+	Positive	63.99
69.	ELISA+	Positive	36.91
70.	ELISA+	Negative	0.06
71.	ELISA+	Negative	0.86
72.	ELISA+	Positive	1.4
73.	ELISA+	Positive	1.45
74.	ELISA+	Positive	27.62
75.	ELISA+	Negative	0.04
76.	ELISA+	Negative	0.06
77.	ELISA+	Positive	20.05
78.	ELISA+	Positive	26.71
79.	ELISA+	Negative	0.13
80.	ELISA+	Positive	49.03
81.	ELISA+	Positive	71.07
82.	ELISA+	Positive	70.68
83.	ELISA+	Negative	0.14
84.	ELISA+	Positive	15.46
85.	ELISA+	Positive	4.58
86.	ELISA+	Negative	0.06
87.	ELISA+	Negative	0
88.	ELISA+	Negative	0.04
89.	ELISA+	Negative	0.03
90.	ELISA+	Positive	29.53
91.	ELISA+	Positive	68.84
92.	ELISA+	Positive	69.2
93.	ELISA+	Negative	0.05
94.	ELISA+	Negative	0.01
95.	ELISA+	Negative	0.02
96.	ELISA+	Positive	15.57
97.	ELISA+	Positive	17.04
98.	ELISA+	Negative	0.01
99.	ELISA+	Positive	71.02
100.	ELISA+	Negative	0.05
101.	ELISA+	Positive	15.06
102.	ELISA+	Positive	3.38
103.	ELISA+	Negative	0.25
104.	ELISA+	Negative	0

105.	ELISA+	Positive	2.19
106.	ELISA+	Positive	69.54
107.	ELISA+	Negative	0.03
108.	ELISA+	Positive	69.1
109.	ELISA+	Negative	0.05
110.	ELISA+	Negative	0.01

ELISA + = ELISA Positive

Table B10. SD Biosensor Standard F™ Dengue NS1- NEG results

	Gold Standard		
	NS1	SD F result	COI
1.	ELISA-	Negative	
2.	ELISA-	Negative	
3.	ELISA-	Negative	
4.	ELISA-	Negative	
5.	ELISA-	Negative	
6.	ELISA-	Negative	
7.	ELISA-	Negative	
8.	ELISA-	Negative	0.01
9.	ELISA-	Negative	0
10.	ELISA-	Negative	0.04
11.	ELISA-	Negative	0.05
12.	ELISA-	Negative	0.05
13.	ELISA-	Negative	0.04
14.	ELISA-	Negative	0.05
15.	ELISA-	Negative	0.02
16.	ELISA-	Negative	0
17.	ELISA-	Negative	0.04
18.	ELISA-	Negative	0.02
19.	ELISA-	Negative	0.13
20.	ELISA-	Negative	0.20
21.	ELISA-	Negative	0
22.	ELISA-	Negative	0.02
23.	ELISA-	Negative	0.04
24.	ELISA-	Negative	0.05
25.	ELISA-	Negative	0
26.	ELISA-	Negative	0
27.	ELISA-	Negative	0.01
28.	ELISA-	Negative	0.06
29.	ELISA-	Negative	0.01
30.	ELISA-	Negative	0.03
31.	ELISA-	Negative	0.04

32.	ELISA-	Negative	0.05
33.	ELISA-	Negative	0.03
34.	ELISA-	Negative	0.01
35.	ELISA-	Negative	0.04
36.	ELISA-	Negative	0.02
37.	ELISA-	Negative	0.05
38.	ELISA-	Negative	0.05
39.	ELISA-	Negative	0
40.	ELISA-	Negative	0.05
41.	ELISA-	Negative	0.06
42.	ELISA-	Negative	0
43.	ELISA-	Positive	10.23
44.	ELISA-	Negative	0.02
45.	ELISA-	Negative	0.05
46.	ELISA-	Negative	0.05
47.	ELISA-	Negative	0
48.	ELISA-	Negative	0
49.	ELISA-	Negative	0.01
50.	ELISA-	Negative	0.04
51.	ELISA-	Negative	0
52.	ELISA-	Negative	0.02
53.	ELISA-	Negative	0.03
54.	ELISA-	Negative	0.03
55.	ELISA-	Negative	0
56.	ELISA-	Negative	0
57.	ELISA-	Negative	0
58.	ELISA-	Negative	0.05
59.	ELISA-	Negative	0.05
60.	ELISA-	Negative	0.01
61.	ELISA-	Negative	0.07
62.	ELISA-	Negative	0.09
63.	ELISA-	Negative	0
64.	ELISA-	Negative	0.04
65.	ELISA-	Negative	0
66.	ELISA-	Negative	0.04
67.	ELISA-	Negative	0
68.	ELISA-	Negative	0
69.	ELISA-	Negative	0.04
70.	ELISA-	Negative	0
71.	ELISA-	Negative	0.06
72.	ELISA-	Negative	0
73.	ELISA-	Negative	0.05
74.	ELISA-	Negative	0.06
75.	ELISA-	Negative	0.06

76.	ELISA-	Negative	0.08
77.	ELISA-	Negative	0
78.	ELISA-	Negative	0.03
79.	ELISA-	Negative	0.02
80.	ELISA-	Negative	0.04
81.	ELISA-	Negative	0.06
82.	ELISA-	Negative	
83.	ELISA-	Negative	0.01
84.	ELISA-	Negative	0.07
85.	ELISA-	Negative	0.03
86.	ELISA-	Negative	0.01
87.	ELISA-	Negative	0.06
88.	ELISA-	Negative	0
89.	ELISA-	Negative	0.10
90.	ELISA-	Negative	0.05
91.	ELISA-	Negative	0.05
92.	ELISA-	Negative	0.03
93.	ELISA-	Negative	0.04
94.	ELISA-	Negative	0.06
95.	ELISA-	Negative	0.02
96.	ELISA-	Negative	0.04
97.	ELISA-	Negative	0.04
98.	ELISA-	Negative	0.06
99.	ELISA-	Negative	0
100.	ELISA-	Negative	0
101.	ELISA-	Negative	0.03
102.	ELISA-	Negative	0
103.	ELISA-	Negative	0.02
104.	ELISA-	Negative	0.11
105.	ELISA-	Negative	0.06
106.	ELISA-	Negative	0
107.	ELISA-	Negative	0
108.	ELISA-	Negative	0.01
109.	ELISA-	Negative	0.03
110.	ELISA-	Negative	0
111.	ELISA-	Negative	0.03
112.	ELISA-	Negative	0
113.	ELISA-	Negative	0.07
114.	ELISA-	Negative	0.05
115.	ELISA-	Negative	0.03
116.	ELISA-	Negative	0.08
117.	ELISA-	Negative	0.04
118.	ELISA-	Negative	0.04
119.	ELISA-	Negative	0.05

120.	ELISA-	Negative	0
121.	ELISA-	Negative	0
122.	ELISA-	Negative	0
123.	ELISA-	Negative	0.02
124.	ELISA-	Negative	0.03

ELISA - = ELISA Negative

Appendix C – Calculations for the positive and negative percent agreement tests for each SD Biosensor Standard™ F FIA test.

Table C1. Sample calculation showing the positive and negative percent agreement tests for SD Biosensor Standard™ F Dengue NS1 FIA tests.

	Type of sample	Investigational device		
		Positive	Negative	Total
Reference method	Positive	70 (a)	40 (b)	(a+b) 110
	Negative	1 (c)	123 (d)	(c+d) 124
	Total	(a+c) 71	(b+d) 163	234

- **Sensitivity** = a/(a+b) x 100%

Positive % agreement = $\frac{a}{(a+b)}$ x 100 = $\frac{70}{110}$ x 100 = **63.63%**

53.92% to 72.60% (95% Confidence Interval)

- **Specificity** = d/(c+d) x 100%

Negative % agreement = $\frac{d}{(c+d)}$ x 100 = $\frac{123}{124}$ x 100= **99.19%**

95.59% to 99.98% (95% Confidence Interval)

- **Overall % agreement** = (a+d)/(a+b+c+d) x 100%
  aka "Accuracy"

Overall % agreement = $\frac{a+d}{(a+b+c+d)}$ x 100 = $\frac{(70+123)}{(70+40+1+123)}$ x100 = **82.47%**

76.99% to 87.12% (95% Confidence Interval)

SD Biosensor Standard™ F Dengue IgM FIA test

An error resulted in insufficient quantities of the SD Biosensor Standard™ F Dengue IgM/IgG FIA Kits being available to complete testing of the DENV IgM samples. As a result, there is no result section for SD Biosensor Standard™ F Dengue IgM FIA test.

Table C2. Sample calculation showing the positive and negative percent agreement tests for SD Biosensor Standard™ F Dengue IgG FIA tests.

	Type of sample	Investigational device		
		Positive	Negative	Total
Reference method	Positive	10 (a)	2 (b)	12 (a+b)
	Negative	0 (c)	1 (d)	1 (c+d)
	Total	10 (a+c)	3 (b+d)	13

- **Sensitivity** = a/(a+b) x 100%

Positive % agreement = $\frac{a}{(a+b)} \times 100 = \frac{10}{12} \times 100 =$ **83.33%**

51.59% to 97.91% (95% Confidence Interval)

- **Specificity** = d/(c+d) x 100%

Negative % agreement = $\frac{d}{(c+d)} \times 100 = \frac{1}{(0+1)} \times 100 =$ **100%**

2.50% to 100% (95% Confidence Interval)

- **Overall % agreement = (a+d)/(a+b+c+d) x 100% aka "Accuracy"**

Overall % agreement = $\frac{a+d}{(a+b+c+d)} \times 100 = \frac{(10+1)}{(10+2+0+1)} \times 100 =$ **84.61%**

54.55% to 98.08% (95% Confidence Interval)

Table C3. Sample calculation showing the positive and negative percent agreement tests for SD Biosensor Standard™ F Zika IgM FIA tests.

	Type of sample	Investigational device		
		Positive	Negative	Total
Reference method	Positive	1 (a)	37 (b)	38 (a+b)
	Negative	0 (c)	111 (d)	111 (c+d)
	Total	1 (a+c)	148 (b+d)	149

- **Sensitivity** = a/(a+b) x 100%

Positive % agreement = $\frac{a}{(a+b)} \times 100 = \frac{1}{38} \times 100 =$ **2.63%**

0.07% to 13.81% (95% Confidence Interval)

- **Specificity** = d/(c+d) x 100%

Negative % agreement = $\frac{d}{(c+d)} \times 100 = \frac{111}{111} \times 100 =$ **100%**

96.73% to 100% (95% Confidence Interval)

- **Overall % agreement = (a+d)/(a+b+c+d) x 100%**
  aka "Accuracy"

Overall % agreement = $\frac{a+d}{(a+b+c+d)} \times 100 = \frac{112}{149} \times 100 =$ **75.17%**

67.43% to 81.87% (95% Confidence Interval)

Table C4. Sample calculation showing the positive and negative percent agreement tests for SD Biosensor Standard™ F Zika IgG FIA tests.

	Type of sample	Investigational device		
		Positive	Negative	Total
Reference method	Positive	0 (a)	6 (b)	6 (a+b)
	Negative	0 (c)	2 (d)	2 (c+d)
	Total	0 (a+c)	8 (b+d)	8

- **Sensitivity** = a/(a+b) x 100%

Positive % agreement = $\frac{a}{(a+b)} \times 100 = \frac{0}{6} \times 100 =$ **0%**

0% to 45.93% (95% Confidence Interval)

- **Specificity** = d/(c+d) x 100%

Negative % agreement = $\frac{d}{(c+d)} \times 100 = \frac{2}{2} \times 100 =$ **100%**

15.81% to 100% (95% Confidence Interval)

- **Overall % agreement = (a+d)/(a+b+c+d) x 100%**
  aka "Accuracy"

Overall % agreement = $\frac{a+d}{(a+b+c+d)} \times 100 = \frac{(0+2)}{(0+6+0+2)} \times 100 =$ **25%**

3.19% to 65.09% (95% Confidence Interval)

Table C5. Sample calculation showing the positive and negative percent agreement tests for SD Biosensor Standard™ F Zika Antigen FIA tests.

	Type of sample	Investigational device		
		Positive	Negative	Total
Reference method	Positive	2 (a)	90 (b)	92 (a+b)
	Negative	1 (c)	100 (d)	101 (c+d)
	Total	3 (a+c)	190 (b+d)	193

- **Sensitivity** = a/(a+b) x 100%

Positive % agreement = $\frac{a}{(a+b)} \times 100 = \frac{2}{92} \times 100 =$ **2.17%**

0.26% to 7.63% (95% Confidence Interval)

- **Specificity** = d/(c+d) x 100%

Negative % agreement = $\frac{d}{(c+d)} \times 100 = \frac{100}{101} \times 100 =$ **99.01%**

94.61% to 99.97% (95% Confidence Interval)

- **Overall % agreement = (a+d)/(a+b+c+d) x 100%**
  aka "Accuracy"

Overall % agreement = $\frac{a+d}{(a+b+c+d)} \times 100 = \frac{(2+100)}{(2+90+1+100)} \times 100 =$ **52.85%**

45.55% to 60.06% (95% Confidence Interval)